

Yenepoya Institute Of Arts Science, Commerce & Management

Project – LOW Level Design

on

Natural Language Interface for Supply Chain Infrastructure

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INTRODUCTION

The Supply Chain Management Chatbot is a command-line based application designed to provide quick access to supply chain information such as orders, shipments, inventory, suppliers, and logistics.

The chatbot uses a rule-based Natural Language Processing approach to interpret user queries and respond with relevant data stored in JSON files.

PROJECT GOAL

The primary goals of the project are:

- To build a chatbot for supply chain queries
- To simplify access to logistics and inventory data
- To demonstrate modular software design
- To simulate real-world SCM operations using structured data

MODEL DEVELOPMENT

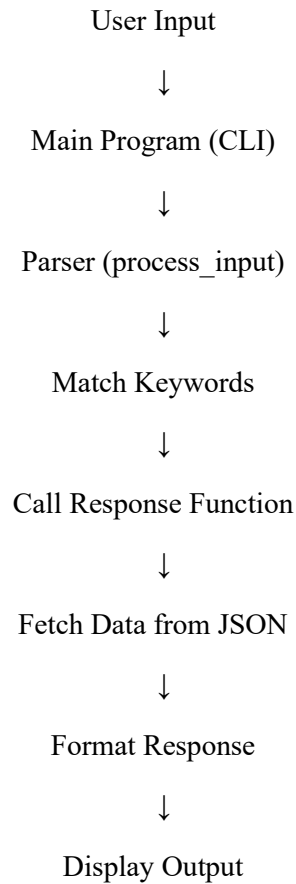
The system is developed using a rule-based model instead of machine learning.

Components:

- Input Processing Module → Captures user input
- Parser Module → Identifies keywords
- Response Module → Fetches and formats data
- Data Layer → JSON-based storage

Logic Used:

- Keyword matching (e.g., "order", "shipment", "inventory")
- Conditional routing using if-elif statements

PROCESS FLOW DIAGRAM

WORKING

- User enters a query (e.g., "order status")
- Input is converted to lowercase
- Parser checks for keywords
- Matching function is called
- Data is retrieved from JSON files
- Output is formatted as readable text
- Response is displayed to the user

Example:

Input:

order status

Output:

Orders:

- Order ID: ORD101 Status: Processing
- Order ID: ORD102 Status: Shipped

DATA DESIGN

Data Source:

All data is stored in JSON files

Orders

- id
- status

Shipments

- tracking
- location

Inventory

- product
- quantity

Products

- product_id
- name
- price

Customers

- customer_id
- name
- city

Suppliers

- name
- product

Warehouses

- warehouse_id
- location
- capacity

Transport

- vehicle_id
- type
- capacity
- driver

CHALLENGES

- Limited understanding due to keyword-based logic
- No real-time data updates
- Dependency on static JSON files
- No error handling for invalid/missing data
- Scalability limitations

REFERENCES

- **google.com**
- **github.com**